11 A planet orbits around a star with a period of T and an orbiting radius of r. The planet is then hit by some asteroids, causing its total energy to decrease and its orbital radius to decrease to $0.25r$.

In this new orbit, how many revolutions does the planet make in the same time T?

A 0.13 **B** 0.25 **C** 4.0 **D** 8.0

Ans: **D**

Gravitational force provides centripetal force

$$\frac{GMm}{r^2} = mr\omega^2 = mr(\frac{2\pi}{T})^2$$

$$T^2 = \frac{4\pi^2}{GM} r^3$$

$$T^2 \propto r^3$$

$$\frac{T_1^2}{T_2^2} = \frac{r_1^3}{r_2^3} = \frac{r^3}{(0.25r)^3} = 64$$

$$T_1 = \sqrt{64} T_2 = 8T_2$$

12 Which statement about geostationary orbits on Earth is **false**?